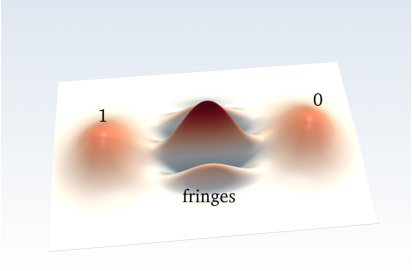
Stabilized cat qubits

DuQuantum 2026, Alice & Bob challenge

Team Shunya

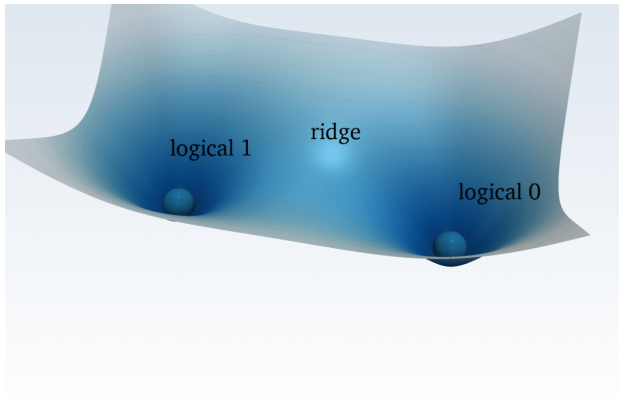


The problem: quantum computers make mistakes

- A qubit can be 0, 1, or both at once, but tiny disturbances from the outside world keep spoiling it
- It goes wrong in two ways: 0 and 1 swap (a **bit flip**), or the “both at once” gets scrambled (a **phase flip**). Fixing both kinds takes many extra qubits

The idea: a cat qubit

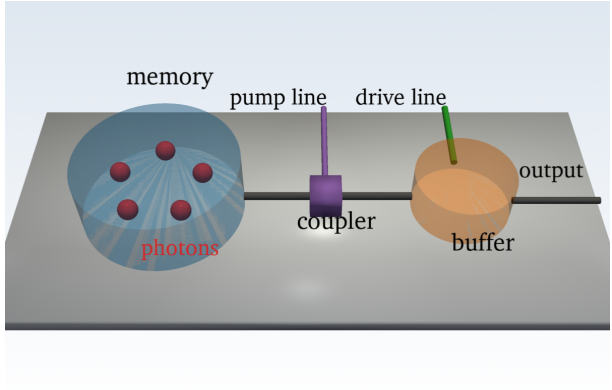
- Store 0 and 1 as a ball resting in one of two valleys, and keep digging the valleys
- Deep valleys mean the ball almost never hops, so bit flips nearly vanish and only phase flips are left to fix



The two valleys, 0 and 1

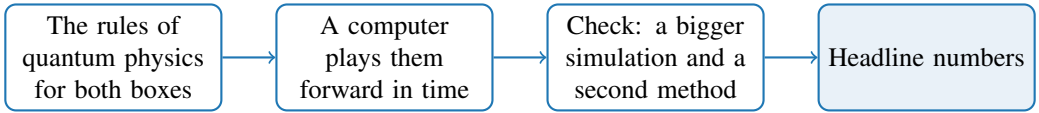
The device we simulated

- A tiny box (the memory) traps microwave light, about 4.4 particles of light (photons). This light is the qubit
- A pump and a leaky helper box add and remove light in pairs, which keeps the two valleys in place



Memory box, pump, helper box and its output line

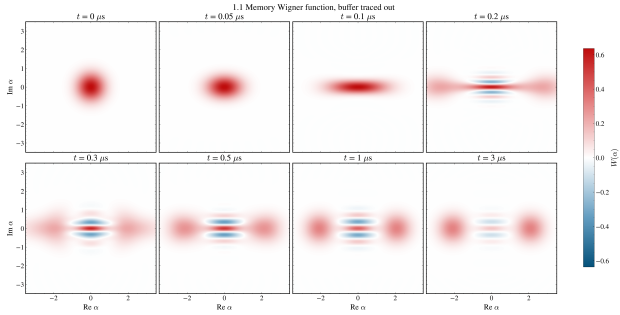
How we solved it: we built the device inside a computer



- We simulated the memory and the helper box together, the way they sit on a real chip
- Every number was checked twice, so the results do not depend on simulation shortcuts

Step 1: the pump turns empty light into a cat

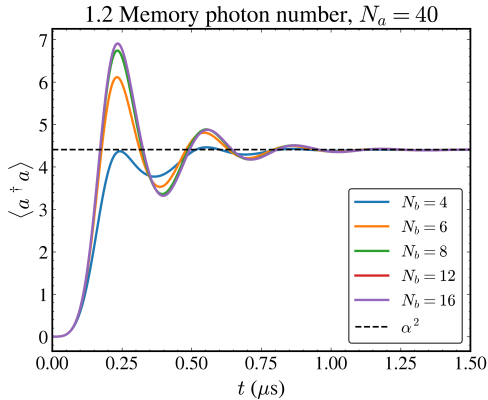
- Snapshots of the light in the box: one blob splits into two, the cat that is 0 and 1 at once
- It takes about 1 microsecond, a millionth of a second



The light in the memory box, from 0 to 3 microseconds

Checking the simulation: the cat settles at 4.4 photons

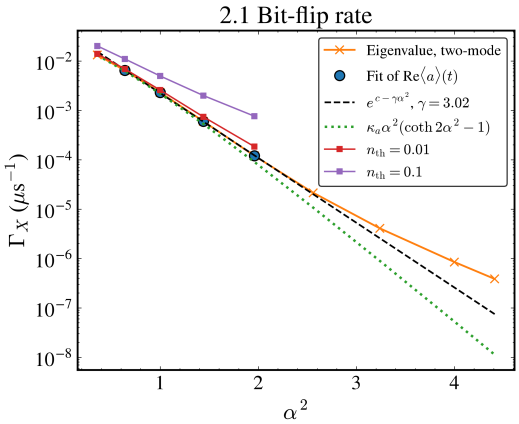
- The light settles at 4.40 photons, the 4.41 the pump settings predict, however big we make the simulation
- The early spike needs more room: the suggested 6 levels for the helper box put it 11.5% too low, so we used 12



Photons in the memory box for different helper-box sizes in the simulation; the dashed line is 4.41

Step 2: bigger cats flip much less

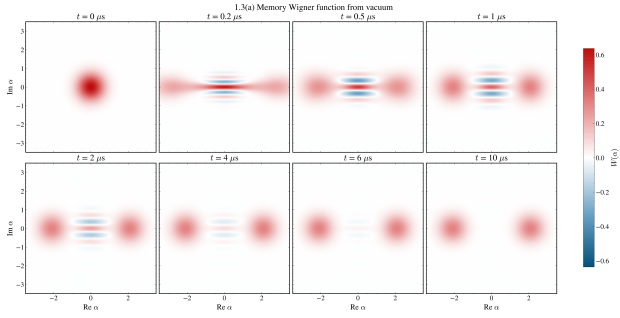
- Each dot is one simulation. Each step down the scale means 10 times fewer bit flips
- For small cats, every extra photon makes bit flips about 20 times rarer



Bit flips per microsecond against cat size

Phase flips: the stripes fade in about 2 microseconds

- The stripes between the blobs show the light is in both valleys at once. Each photon that leaks out of the box weakens them
- We timed the fading: a phase flip every 1.94 microseconds on average, as the textbook formula predicts



The light in the memory box from 0 to 10 microseconds: the blobs stay, the stripes fade

The big result: bit flips are a million times rarer

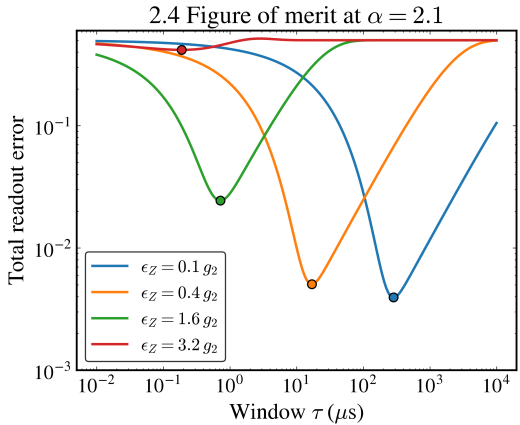
2.6 s	1.94 μs	1,000,000$\times$
average time to a bit flip	average time to a phase flip	how much rarer bit flips are

- At our working size, a cat of 4.4 photons
- If phase flips happened once a second, bit flips would happen about once every two weeks

Step 3: reading the qubit has a sweet spot

- A gentle push makes the helper box's signal go up for 0 and down for 1. Read too briefly and the signal is too faint; read too long and the cat may flip while we look

- Each push strength has a best reading time (the dots). The best is 0.4% wrong answers, with the gentlest push read for 285 microseconds



Readout mistakes against reading time, for four push strengths

Step 4: a bigger cat while reading cuts mistakes 13 to 18 times

$0.41\% \rightarrow 0.023\%$

gentle push: 18 times fewer mistakes

$0.87\% \rightarrow 0.066\%$

stronger push: 13 times fewer mistakes

- We grow the cat from 4.4 to 6.25 photons only while reading, then shrink it back
- A bigger cat is harder to knock over, so the push flips it far less. Reading too long still adds mistakes, just much later

What we found

- The cat qubit keeps its bit for seconds: bit flips are a million times rarer than phase flips, so correcting errors gets much cheaper
- It can be read with 0.4% wrong answers, and 13 to 18 times fewer when the cat is made bigger during the reading

Code and every number: github.com/yubrajbhandari923/DuQuantum, branch ak-changes